

SERVE: margin-aware cache-reuse pipeline

Incoming query q

Top k nearest-neighbour search in cache

$k = 8$ nearest entries

Four zero-cost retrieval byproducts
 $s_1, m = s_1 - s_2, s_2, \rho$ (**local density**)

discarded by GPTCache / MeanCache / vCache

Gradient-boosted gate (GBDG)

4-dim input: $[s_1, m, s_2, \rho]$
150 trees, max depth 13, 0.8 us/query

Margin $m = s_1 - s_2$ alone
recovers 83% of the total
improvement over a
similarity-only gate

Reuse cached answer

Call LLM